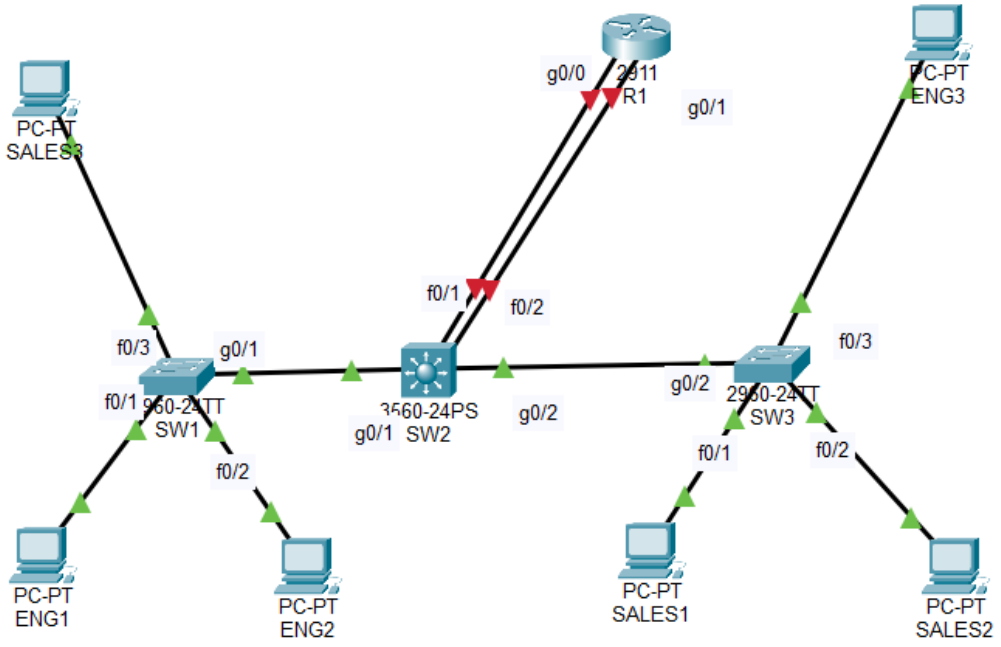
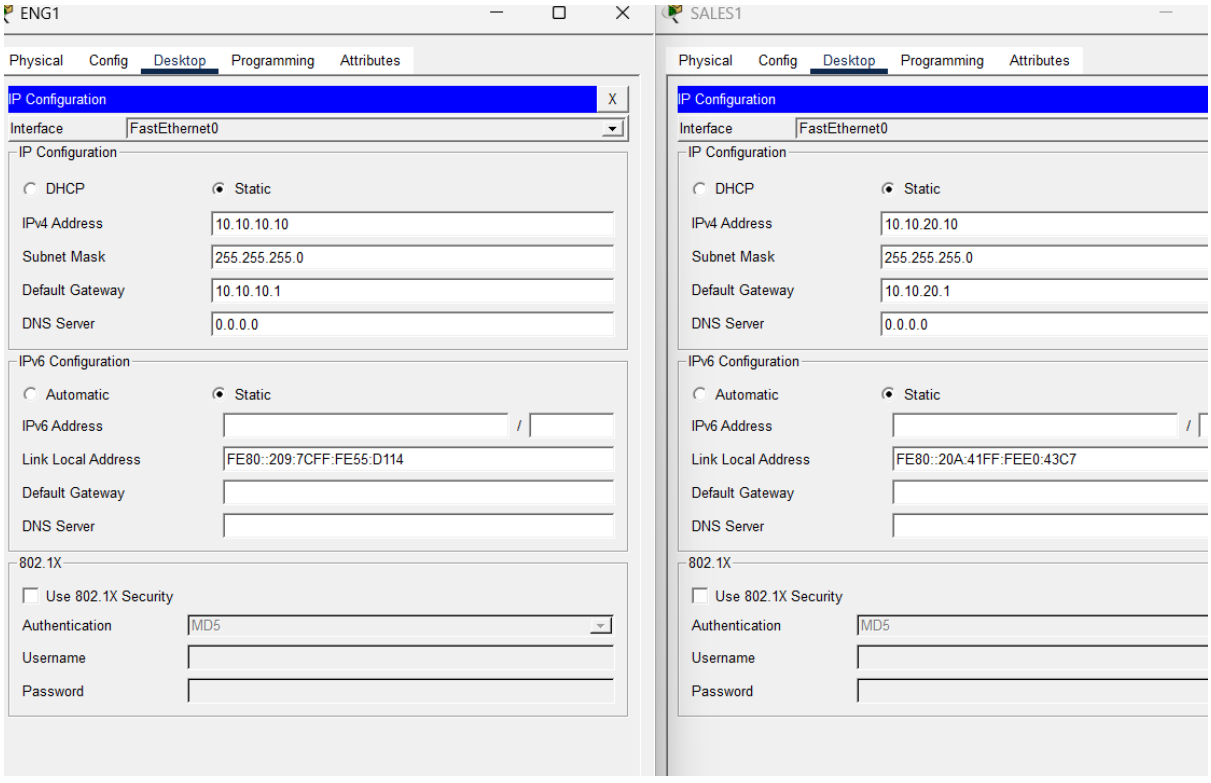
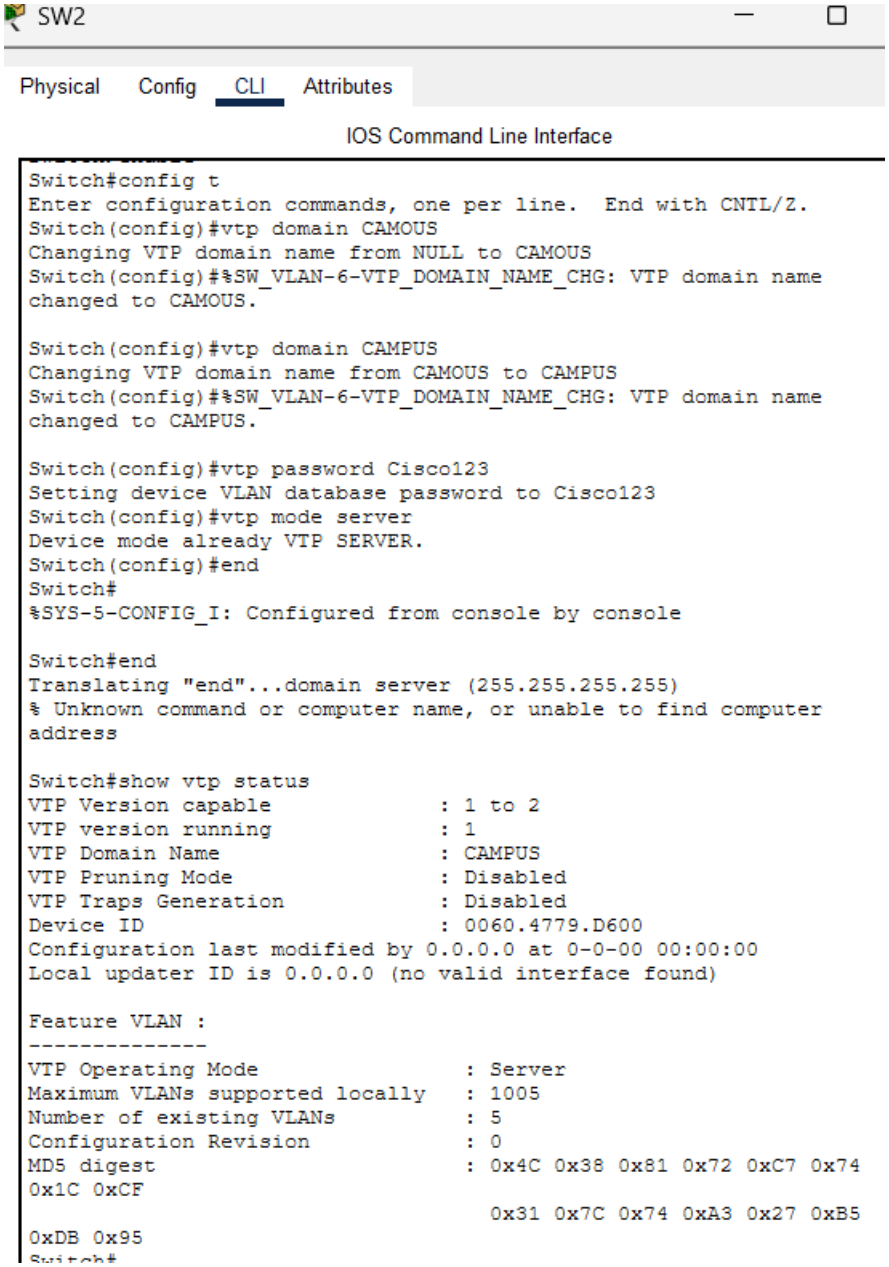
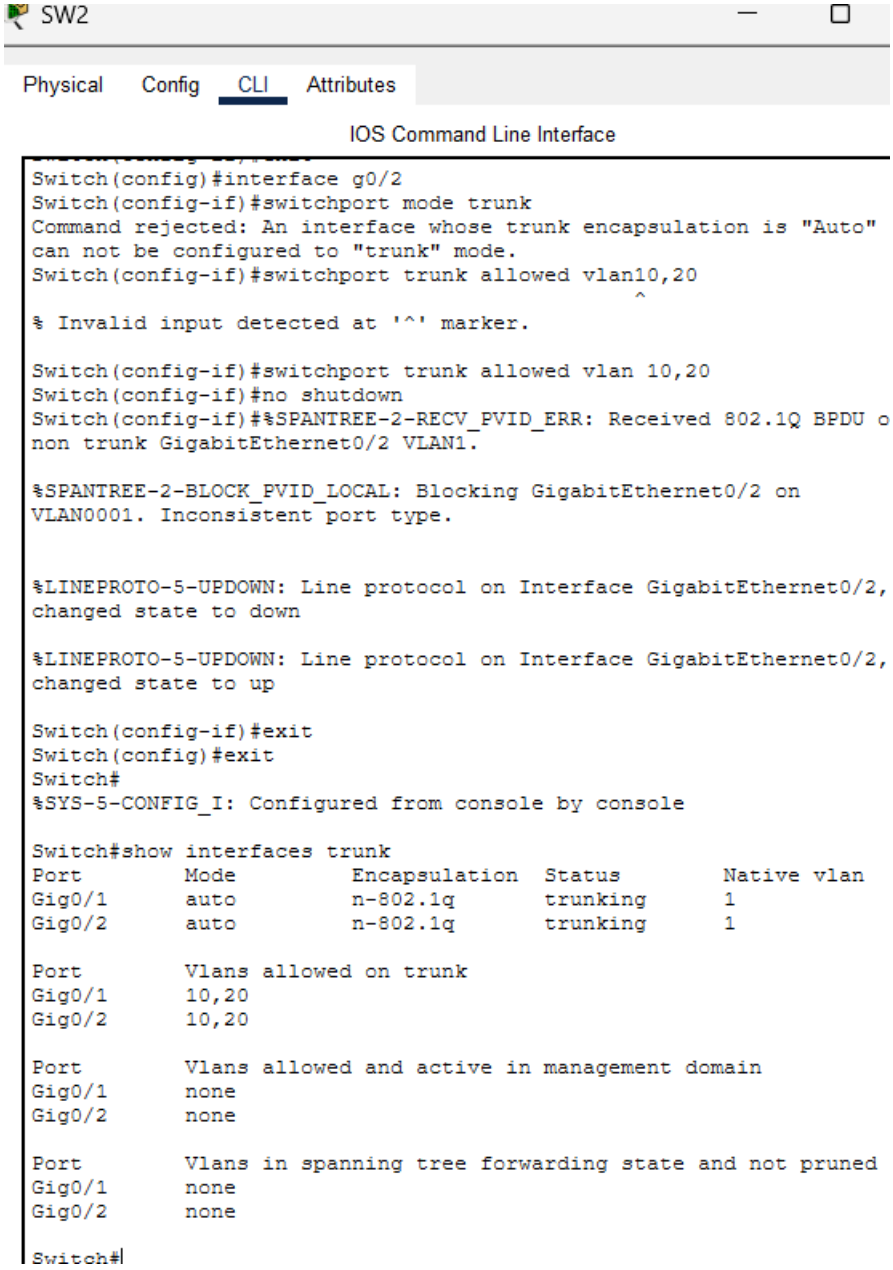

PC-PT
SALES3
g0/0
2911
R1
g0/1
PC-PT
ENG3
f0/1
f0/2
f0/3
g0/1
f0/1
960-24TT
SW1
3560-24PS
g0/1
SW2
g0/2
g0/2
2960-24TT
SW3
f0/3
f0/1
f0/2
f0/2
PC-PT
ENG1
PC-PT
ENG2
PC-PT
SALES1
PC-PT
SALES2


ENG1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.10.10.10
Subnet Mask
255.255.255.0
Default Gateway
10.10.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::209:7CFF:FE55:D114
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
SALES1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.10.20.10
Subnet Mask
255.255.255.0
Default Gateway
10.10.20.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20A:41FF:FEE0:43C7
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


SW2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vtp domain CAMOUS
Changing VTP domain name from NULL to CAMOUS
Switch(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name
changed to CAMOUS.
Switch(config)#vtp domain CAMPUS
Changing VTP domain name from CAMOUS to CAMPUS
Switch(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name
changed to CAMPUS.
Switch(config)#vtp password Cisco123
Setting device VLAN database password to Cisco123
Switch(config)#vtp mode server
Device mode already VTP SERVER.
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer
address
Switch#show vtp status
VTP Version capable   : 1 to 2
VTP version running   : 1
VTP Domain Name   : CAMPUS
VTP Pruning Mode   : Disabled
VTP Traps Generation   : Disabled
Device ID   : 0060.4779.D600
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)
Feature VLAN :

VTP Operating Mode   : Server
Maximum VLANs supported locally   : 1005
Number of existing VLANs   : 5
Configuration Revision   : 0
MD5 digest   : 0x4C 0x38 0x81 0x72 0xC7 0x74
0x1C 0xCF
                                    0x31 0x7C 0x74 0xA3 0x27 0xB5
0xDB 0x95


SW2
Physical
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IOS Command Line Interface
Switch(config)#interface g0/2
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto"
can not be configured to "trunk" mode.
Switch(config-if)#switchport trunk allowed vlan10,20
^
% Invalid input detected at '^' marker.
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#no shutdown
Switch(config-if)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU o
non trunk GigabitEthernet0/2 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/2 on
VLAN0001. Inconsistent port type.
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show interfaces trunk
Port Mode Encapsulation Status Native vlan
Gig0/1 auto n-802.1q trunking 1
Gig0/2 auto n-802.1q trunking 1
Port Vlans allowed on trunk
Gig0/1 10,20
Gig0/2 10,20
Port Vlans allowed and active in management domain
Gig0/1 none
Gig0/2 none
Port Vlans in spanning tree forwarding state and not pruned
Gig0/1 none
Gig0/2 none
Switch#

Physical Config CLI Attributes

IOS Command Line Interface

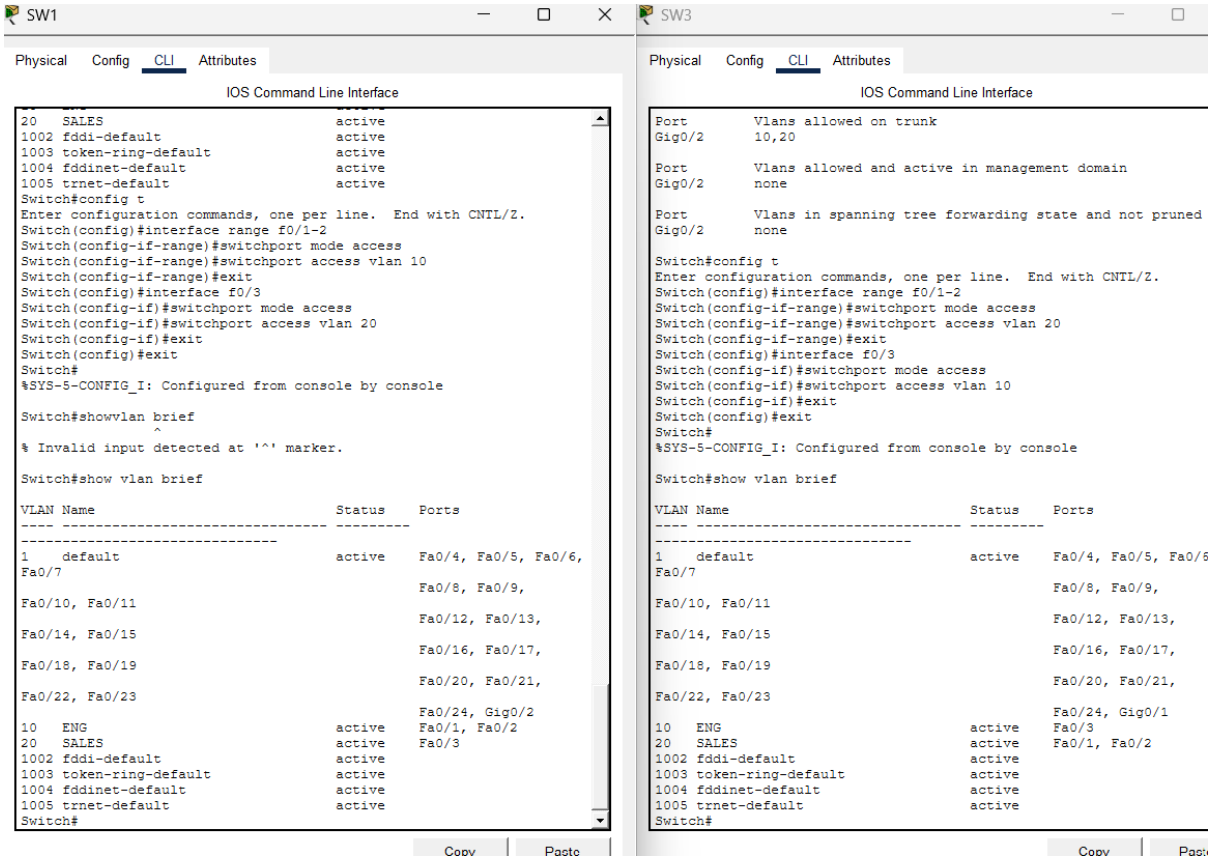
```
Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      none
Gig0/2      none

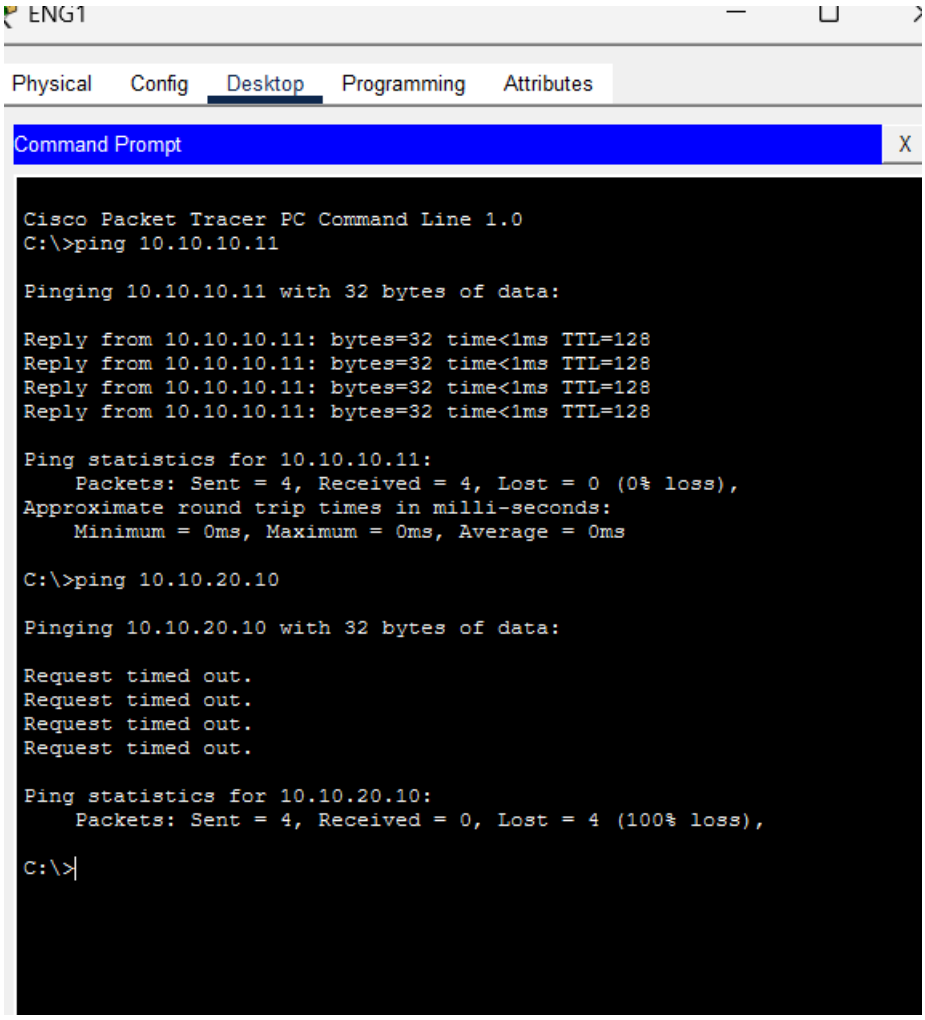
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name ENG
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name SALES
Switch(config-vlan)#exit
Switch(config)#show vlan brief
                ^
% Invalid input detected at '^' marker.

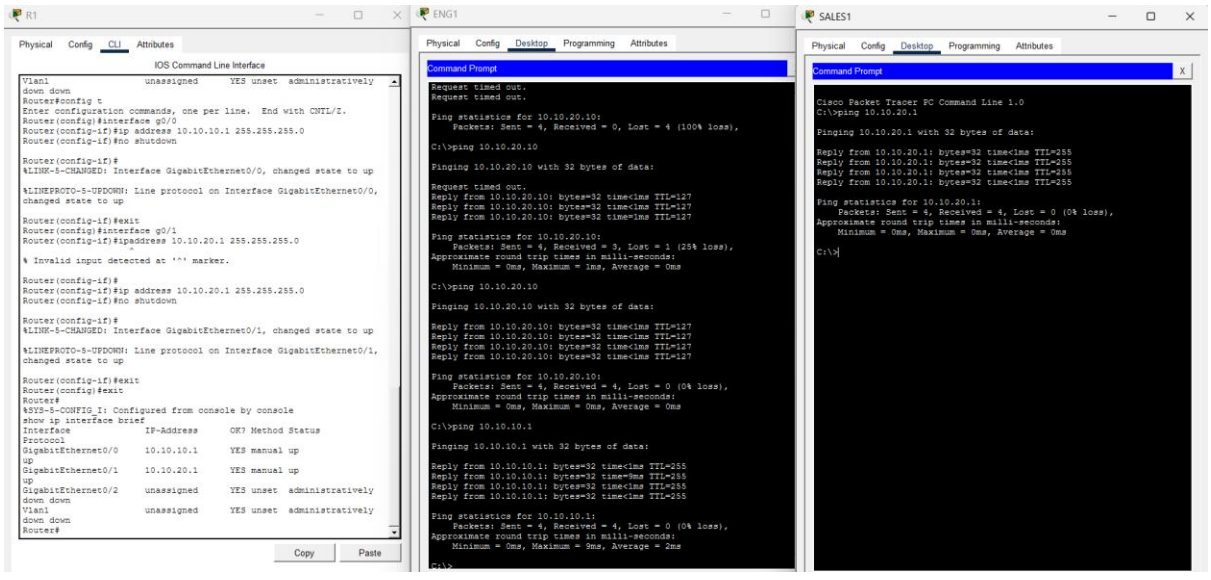
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
sh
% Incomplete command.
Switch#
Switch#show vlan brief

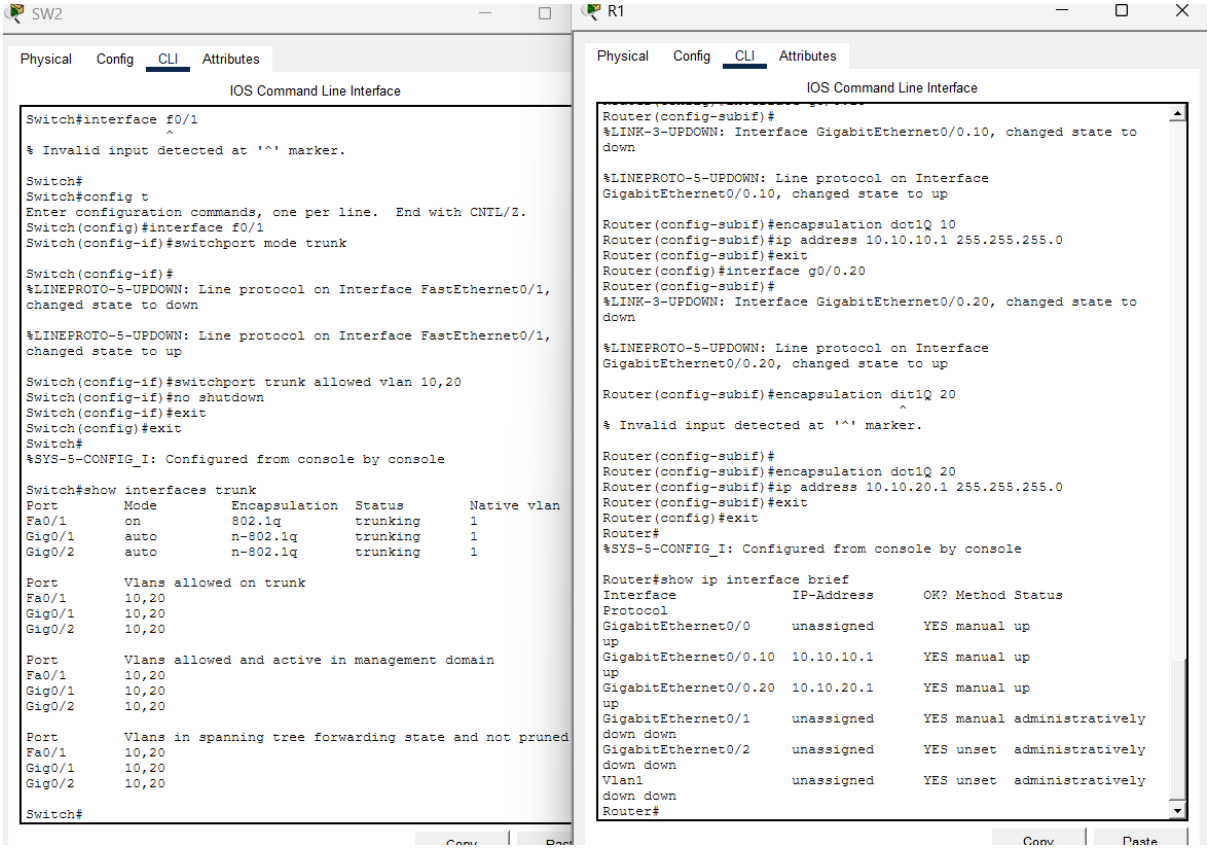
VLAN Name                              Status    Ports
---- --------------------------------- ---------
--------------------------------
1    default                           active    Fa0/1, Fa0/2, Fa0/3,
Fa0/4
                                                 Fa0/5, Fa0/6, Fa0/7,
Fa0/8
                                                 Fa0/9, Fa0/10,
Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14,
Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18,
Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22,
Fa0/23, Fa0/24
10   ENG                               active
20   SALES                             active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#
```

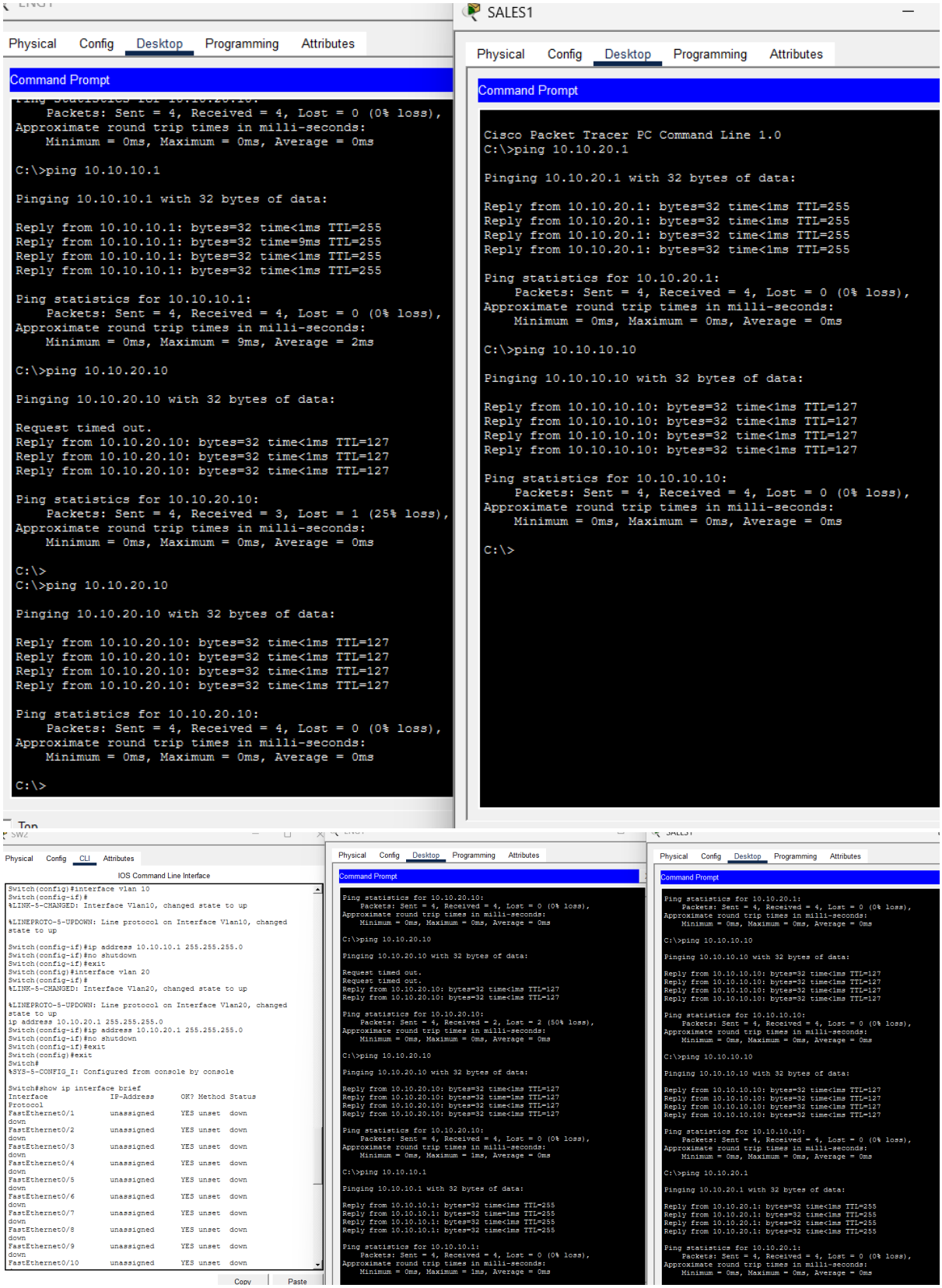
Copy Paste

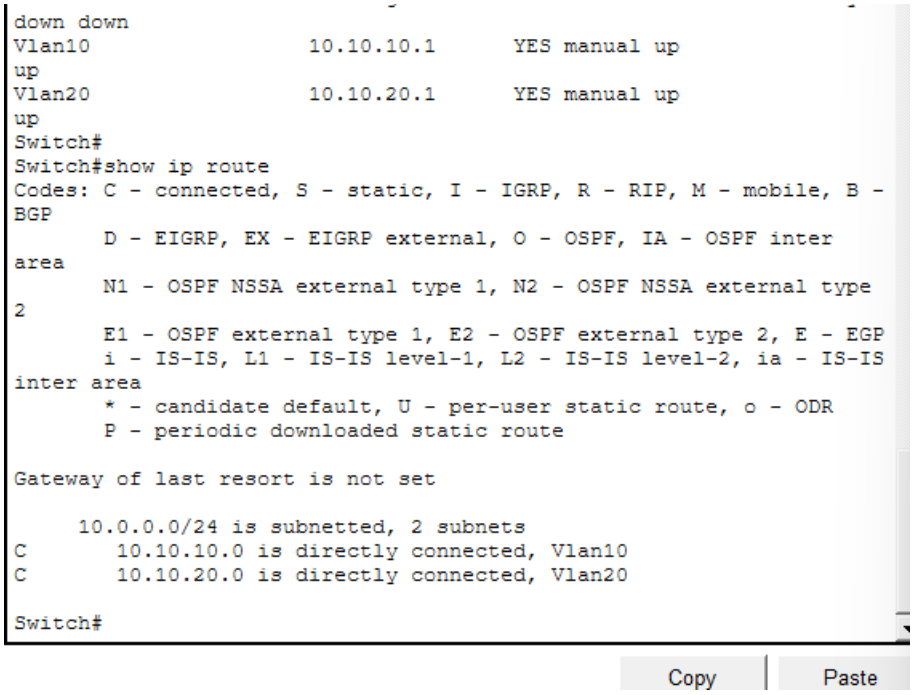

SW1
Physical
Config
CLI
Attributes
IOS Command Line Interface
20 SALES active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range f0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface f0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#showvlan brief
^
% Invalid input detected at '^' marker.
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/4, Fa0/5, Fa0/6, Fa0/7
Fa0/8, Fa0/9, Fa0/10, Fa0/11
Fa0/12, Fa0/13, Fa0/14, Fa0/15
Fa0/16, Fa0/17, Fa0/18, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/2
10 ENG active Fa0/1, Fa0/2
20 SALES active Fa0/3
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
Copy
Paste
SW3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Port Vlans allowed on trunk
Gig0/2 10,20
Port Vlans allowed and active in management domain
Gig0/2 none
Port Vlans in spanning tree forwarding state and not pruned
Gig0/2 none
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range f0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface f0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief
VLAN Name Status Ports
1 default active Fa0/4, Fa0/5, Fa0/6, Fa0/7
Fa0/8, Fa0/9, Fa0/10, Fa0/11
Fa0/12, Fa0/13, Fa0/14, Fa0/15
Fa0/16, Fa0/17, Fa0/18, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/1
10 ENG active Fa0/3
20 SALES active Fa0/1, Fa0/2
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
Copy


ENG1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.11
Pinging 10.10.10.11 with 32 bytes of data:
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Ping statistics for 10.10.10.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.20.10
Pinging 10.10.20.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 10.10.20.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>


R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Copy
Paste
ENG1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
SALES1
Physical
Config
Desktop
Programming
Attributes
Command Prompt


SW2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#interface f0/1
^
% Invalid input detected at '^' marker.
Switch#
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface f0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show interfaces trunk
Port Mode Encapsulation Status Native vlan
Fa0/1 on 802.1q trunking 1
Gig0/1 auto n-802.1q trunking 1
Gig0/2 auto n-802.1q trunking 1
Port Vlans allowed on trunk
Fa0/1 10,20
Gig0/1 10,20
Gig0/2 10,20
Port Vlans allowed and active in management domain
Fa0/1 10,20
Gig0/1 10,20
Gig0/2 10,20
Port Vlans in spanning tree forwarding state and not pruned
Fa0/1 10,20
Gig0/1 10,20
Gig0/2 10,20
Switch#
R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to
down
%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0.10, changed state to up
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 10.10.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to
down
%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0.20, changed state to up
Router(config-subif)#encapsulation dit1Q 20
^
% Invalid input detected at '^' marker.
Router(config-subif)#
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 10.10.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface IP-Address OK? Method Status
Protocol
GigabitEthernet0/0 unassigned YES manual up
up
GigabitEthernet0/0.10 10.10.10.1 YES manual up
up
GigabitEthernet0/0.20 10.10.20.1 YES manual up
up
GigabitEthernet0/1 unassigned YES manual administratively
down down
GigabitEthernet0/2 unassigned YES unset administratively
down down
Vlan1 unassigned YES unset administratively
down down
Router#


Physical
Config
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Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.10.1
Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=9ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 9ms, Average = 2ms
C:\>ping 10.10.20.10
Pinging 10.10.20.10 with 32 bytes of data:
Request timed out.
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.20.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
C:\>ping 10.10.20.10
Pinging 10.10.20.10 with 32 bytes of data:
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.20.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
SALES1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.20.1
Pinging 10.10.20.1 with 32 bytes of data:
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.20.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.10.10
Pinging 10.10.10.10 with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
Switch(config-if)#ip address 10.10.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
ip address 10.10.20.1 255.255.255.0
Switch(config-if)#ip address 10.10.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/1 unassigned YES unset down down
FastEthernet0/2 unassigned YES unset down down
FastEthernet0/3 unassigned YES unset down down
FastEthernet0/4 unassigned YES unset down down
FastEthernet0/5 unassigned YES unset down down
FastEthernet0/6 unassigned YES unset down down
FastEthernet0/7 unassigned YES unset down down
FastEthernet0/8 unassigned YES unset down down
FastEthernet0/9 unassigned YES unset down down
FastEthernet0/10 unassigned YES unset down
Copy
Paste
Physical
Config
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Command Prompt
Ping statistics for 10.10.20.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.20.10
Pinging 10.10.20.10 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.20.10:
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.20.10
Pinging 10.10.20.10 with 32 bytes of data:
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.20.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 10.10.10.1
Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
Physical
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Command Prompt
Ping statistics for 10.10.20.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.10.10
Pinging 10.10.10.10 with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.10.10
Pinging 10.10.10.10 with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Reply from 10.10.10.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.20.1
Pinging 10.10.20.1 with 32 bytes of data:
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.20.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms



The Layer 3 switch method (Option 3) would scale best in a large campus network. This is because routing takes place directly on the multilayer switch using Switched Virtual Interfaces (SVIs), instead of sending traffic to an external router. This provides faster communication between VLANs, reduces the need for multiple physical router connections and simplifies network management. It is also suitable for large networks with many VLANs because routing can be handled efficiently by the switch.